

Complete Voice Assistant Module for Eye Tracking Virtual Mouse System

Overview

This module enables hands-free typing using speech recognition. Users can speak complete sentences or individual keys. It supports accessibility and productivity across desktop applications.

Applications

Microsoft Word: documents, reports, assignments.

Microsoft Excel: numbers, formulas, navigation.

Microsoft PowerPoint: presentations, slides and notes.

Keyboard Support

Letters A-Z; Numbers 0-9; Function Keys F1-F12; Control Keys: Space, Enter, Backspace, Delete, Tab, Caps Lock, Shift, Ctrl, Alt, Esc, Windows, Menu; Navigation: Up, Down, Left, Right, Home, End, Page Up, Page Down, Insert, Print Screen, Scroll Lock, Pause/Break; Numeric Keypad: Num Lock, 0-9, +, -, *, /, Decimal, Enter; Symbols: ! @ # \$ % ^ & * () - _ = + [] { } \ | ; : ' " , . / ? < >

Voice Commands

Single-key mode: 'A', 'Enter', 'Space', etc.

Sentence mode: Speak naturally for speech-to-text.

Optional: Copy, Paste, Cut, Undo, Redo, Save, Select All.

Working Flow

1. Activate assistant.
2. Capture voice.
3. Speech recognition.
4. Command processing.
5. Virtual keystrokes.
6. Output in active app.

Keyboard Mode

Voice Command: **Open Keyboard**

Displays full QWERTY keyboard for key-by-key typing.

Number Mode

Number Mode On: activates numeric keypad (Zero-Nine).

Number Mode Off: returns to normal mode.

Symbol Mode

Voice Command: **Symbol Mode**. Supports At, Hash, Dollar, Percent, Plus, Minus, Slash, Equal, Colon, Semicolon, Comma, Full Stop.

Capital Letter Mode

Caps On: uppercase typing.

Caps Off: lowercase typing.

Navigation Mode

Move Up, Move Down, Move Left, Move Right, Next Line, Previous Line, Home, End.

Editing Mode

Delete Character, Delete Word, Delete Line, Undo, Redo, Select All, Copy, Cut, Paste.

Microsoft Word Mode

Word Mode: New Paragraph, Bold, Italic, Underline, Increase/Decrease Font Size, Save Document.

Microsoft Excel Mode

Excel Mode: Next Cell, Previous Cell, Row Up/Down, New Column, Sum, Average, Save Sheet.

Microsoft PowerPoint Mode

Presentation Mode: New Slide, Next/Previous Slide, Start/End Slideshow, Add Title, Add Bullet Point, Save Presentation.

Browser Mode

Open Google, Search, Scroll Up/Down, Refresh, Close Tab.

Eye Tracking + Voice

User controls cursor with eye tracking and gives voice commands such as Start Dictation, New Paragraph, Bold, Save Document.

AI Smart Commands

Correct Grammar, Translate to English, Translate to Urdu, Read Selected Text, Stop Reading, Increase Volume, Decrease Volume.

Advantages

Hands-free operation, accessibility, higher productivity, support for disabled users, Office integration, AI-powered computer interaction.